As I have pounded into you to this point, systems are systems of systems.

It is reasonable to expect that common planning categories exist for all systems (and systems of systems).

The Nine Categories

(1) Function (the reason the system, or subsystem, exists)

(2) Functional Implementation (specifics on making (1) work that can be in the form of requirements or constraints)

what is the difference?

(3) Interface (External or inter-modular, inputs and outputs)

(4) Standards and Regulations (In-house, industry, or government things that must be met)

(5) Safety

(6) Energy

(7) Environments (under what conditions must the device function, both inactive and active)

storage temp. comes to mind

(8) Structure (how must it be contained, mechanical requirements etc.)

(9) Misc. (category due to my cowardice)

Nine Basic Categories of Requirements (The 9 Questions)

(1) Function (reason for existence)

(2) Functional Implementation (requirements or constraints for achieving (1))

(3) Interface (external or inter-modular)

(4) Standards and Regulations

(5) Safety

(6) Energy

(7) Environments

(8) Structure

(9) Misc.

Inter-Modular Interface Rules

- If a module produces an output, that output is defined in the module.
- If a module uses an input, the input is referenced to its defining module.

Document Structure

System Level

Go through the 9-categories and write down requirements that apply to the whole system

Module 1

Go through the 9 categories and write down requirements unique to this module

⋮

Module n

External Interface Rule

The system (or module) defines what it requires of an input and what it produces as an output.